

Steve Gwade

Fullstack AI Engineer

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Professional Summary

Fullstack AI Engineer with 2.5 years of cumulated experience building and deploying web applications integrated with machine learning models. Develops asynchronous server architectures (FastAPI, Django), type-safe frontends (React, TypeScript, Vite), and model orchestration layers (LLM pipelines, multi-modal applications, and vector search indices). Experienced engineering background event streams, writing automated test suites, and managing containerized runtime environments for low-latency production applications.

Technical Skills

- **Languages:** Python (Asyncio), TypeScript, JavaScript, Kotlin, SQL (PostgreSQL, SQLite)
- **Protocols:** WebSockets, REST, Bash.
- **Frontend:** React, Vite assembly toolchains, Next.js, Redux, Tailwind CSS, Frontend State Management.
- **Backend & Infrastructure:** FastAPI, Django, Nginx, Redis Caching layers, Asynchronous Task Queues, Microservices.
- **AI Engineering:** LLMs, Multi-Modal Systems, LangChain, Structured Data Parsing (Pydantic validation).
- **Machine Learning:** PyTorch, Vector Search Systems (FAISS), TensorRT INT8 Edge Calibration, CUDA.
- **DevOps & Observability:** Docker Compose, GitHub Actions CI/CD, Hetzner, Caddy, Cloudflare, n8n, AWS (SQS, EC2, S3), MLflow.
- **Methodologies:** Agile/Scrum Sprint Lifecycles, Unit & Integration Testing, API Contract Verification.

Work Experience

- **Software Engineer** (*Full-time*) Jul 2026 – Present
Sawego Digital Yaounde, Cameroon
 - Helped move 45M+ FCFA in customer deposits across 2 telecom operators by contributing across the stack of Nexus, a transaction platform: FastAPI backend (72 endpoints), React console, Android client, PostgreSQL and infrastructure.
 - Kept customer funds free of duplicates and false matches, guarded by 735 automated tests, by implementing idempotency and correlation logic that routes ambiguous payments to manual review.
 - Shipped 11 signed Android releases through health-check gated GitHub Actions pipelines on Hetzner (Docker, Caddy).
 - Gave a remittance team an early churn signal, reaching 0.816 AUC over 6 walk-forward windows, by building Vigie's backend, PostgreSQL database with row level security and infrastructure around a survival model.
 - Co-built a bilingual support assistant (n8n) with a teammate, adding Redis debouncing, human handoff and 77 tests.
- **Fullstack AI Engineer** (*Freelance*) Mar 2026 – Jul 2026
MITS SARL – Moabi B2B Travel Agency Yaounde, Cameroon
 - Built an asynchronous LLM orchestration layer using FastAPI and structured Pydantic validation to handle real-time WhatsApp conversational streams, reducing 99th-percentile API response latency to under 120ms.
 - Implemented responsive, multi-turn chat views in React and TypeScript with localized context management, backed by a distributed Redis and DynamoDB caching layer handling 500+ concurrent user streams.
 - Decoupled heavy model inference components from core application web servers using event-driven AWS EventBridge and SQS queues, reducing cloud resource provisioning costs by 35%.
 - Mentored 2 junior engineers while setting up unified GitHub Actions CI/CD workflows to execute parallel PyTest and Vitest suites, maintaining over 92% test coverage across 15+ core system modules.
- **Fullstack AI Solutions Engineer** (*Part-time*) Jan 2025 – Jan 2026
JIANTS – Systems Intelligence Lab Douala, Cameroon
 - Deployed 4 multi-modal pipelines (Vision, OCR, STT, TTS) inside React/TypeScript evaluation dashboards to expose real-time model metrics to enterprise clients.
 - Refactored PyTorch tensor operations and data-loading layers using MLflow tracking, reducing pipeline feature ingestion bottlenecks by 25%.
 - Quantized local computer vision layers via TensorRT INT8 post-training protocols, boosting model inference speed from 18 to 28 FPS on embedded CUDA platforms.
 - Audited runtime memory footprints to implement tensor activation checkpointing, reclaiming 40% VRAM during concurrent batch execution cycles.
- **Fullstack Software Engineer** (*Part-time*) Mar 2024 – Dec 2024
TGNix – Information & Technology Consulting Douala, Cameroon
 - Integrated predictive modules into 2 ERP web applications using React, reduced administrative review times by 30%.
 - Reduced SQL execution times by 25% across 10M+ transaction records via query optimization and composite indexing.
 - Engineered non-blocking FastAPI backend routes for distributed calculation jobs, handling 150k+ daily API updates without message dropping or connection timeouts.

- Containerized multi-tiered application environments using Docker and Docker Compose, reducing local developer onboarding spin-up times from 4 hours to 15 minutes.
- **AI Backend Engineer** (*Internship*) Jun 2023 – Nov 2023
AITECAF – AI Technologies for Africa Yaounde, Cameroon
 - Designed and implemented RESTful Python APIs serving 5 internal classification models, maintaining 99.95% system uptime across a 6-month tracking window.
 - Designed automated text parsing and embedding generation layers using LangChain, scaling document extraction infrastructure to process 50k+ tokens per second.
 - Fine-tuned a custom YOLOv8 object detection model architecture, optimizing anchor box allocations to achieve a sub-5ms processing speed and a 4.2% increase in mAP@0.5.
 - Automated data engineering ingestion pipelines for 10k+ high-resolution images, using stratified spatial hashing to eliminate evaluation data leakage.

Selected Projects

Kite, Cross-Device Scratchpad (*Open Source, 2026*)

- Architected and built a cross-device collaborative workspace web application using React (Vite, TypeScript) and an asynchronous FastAPI WebSocket backend, enabling real-time synchronization across desktop and mobile clients.
- Designed an offline-first data reconciliation mechanism using client-side state diffing, maintaining document parity for 1,200+ active concurrent sessions.
- Implemented batched REST synchronization endpoints, reducing total network payloads by 20% during peak usage periods.
- Engineered connection recovery and automatic state resynchronization workflows, reducing manual refreshes after network interruptions by over 90%.

VERO, Workspace for Grounded AI (*Open-Source Project, 2025*)

- Engineered a document-search RAG platform using a React User Interface, indexing 100k+ multi-format document blocks with context retrieval responses under 45ms.
- Configured a hybrid search pipeline combining dense vector embeddings with BM25 keyword tracking, optimized using Reciprocal Rank Fusion (RRF).

LINA, Assistive Multi-Modal AI Glasses for the Blind (*Huawei ICT Competition 2025*)

- Co-developed a multi-modal assistive system integrating Raspberry Pi 4, camera, audio, and sensor streams with real-time AI inference and WebSocket-based communication.
- Trained and integrated anti-spoofing classifiers to secure access control, successfully blocking 99% of visual spoofing attacks.
- Improved user mobility by contributing to an AI pipeline that increased navigation confidence by 90% while reducing navigation assistance needs by 75% during prototype trials.
- 1st Place Worldwide (Innovation Track) at Huawei ICT Competition 2025 among 179 finalist teams from 48 countries.

PlumVision, Computer Vision Quality Control System (*JCIA Challenge Winner 2025*)

- Trained and deployed a $\approx 3\text{MB}$ ConvNeXt image classification service using representation learning, reaching a 98% accuracy rating across production test samples.
- Wrote a monitoring web interface using React and FastAPI to stream analytics, completing frame execution cycles safely within a 12ms target window.

Academic Qualifications

- **MSc in Engineering (Data Science & Artificial Intelligence)** 2023 – 2025
National Higher Polytechnic School of Douala Douala, Cameroon
 - Graduation Status: Graduated with Distinction (Top 3% of cohort, **GPA: 3.7/4**).
 - Core Focus: Distributed Systems Architecture, Applied Deep Learning Engineering, High-Scale Data Processing Pipelines.
 - **MSc Source Code:** github.com/GwadeSteve/MTTL-Research-Malaria
- **BSc in Engineering (Computer Science)** 2020 – 2023
National Higher Polytechnic School of Douala Douala, Cameroon
 - Top 5% of cohort, **GPA: 3.5/4**.
 - Core Modules: Object-Oriented Design, Problem Solving, Algorithmic Complexity analysis, Advanced Data Structures, Relational Database Engineering.

Awards and Recognition

- **1st Place Worldwide Winner**, Huawei ICT Competition 2025 (Innovative Track) for the LINA Assistive Vision project.
- **Ranked 2nd Across Africa**, ODC 2025, Algorithmic and Competitive Programming tournament (CodinGame).
- **Winner, Best AI Solution** for Optimization and Efficiency, JCIA 2025 Track for PlumVision.
- **Winner, Most Innovative AI Prototype**, JCIA 2025 Track Awarded for EatWise, a multi-modal framework for targeted nutritional education.